

Parv Patel

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Education

- Northeastern University**, Master of Science in Software Engineering Systems May 2025
- GPA: 3.7/4.0
 - **Coursework:** Web Development, Data Management & Database Design, Algorithms, Object-Oriented Design
- Pandit Deendayal Energy University**, Bachelor of Technology in Information and Communication Technology June 2023
- GPA: 9.18/10.0

Experience

- Software Engineering Intern**, Monarch Innovation Private LTD – Guj, India May 2023
- Built and optimized front-end architecture for a cross-platform 3D visualization tool using Express.js and Three.js, improving real-time rendering performance.
 - Developed advanced 3D rendering components and streamlined pipelines, cutting client-side computation by 40% and delivering smoother user interactions.
 - Collaborated with controls engineers, product managers, and QA teams to translate operational requirements into scalable, fault-tolerant backend services.

Projects

- 3D-Bin-Packing – Real-time Packing Visualizer** Jan 2025
[doi: 10.1109/ASIANCON58793.2023.10270781](https://doi.org/10.1109/ASIANCON58793.2023.10270781) | [Github Repo](#)
- Implemented a 3D bin packing algorithm in TypeScript based on the Air Force “human intelligence-based” approach, optimizing the arrangement of box sets within constrained pallet volumes.
 - Designed a skyline-based spatial indexing system with layer heuristics, boosting packing efficiency by 18% over baseline greedy strategies.
 - Integrated Node.js for algorithm execution and testing, and Three.js for 3D visualization of packing layouts, enabling debugging and interactive analysis of space utilization.
 - Published results at IEEE ASIANCON 2023, showcasing performance gains to the research community.
- Spotlight - Human Rights Violation Reporting Platform** | [Github Repo](#) Mar 2024
- Engineered a full-stack, web-based platform using React, TypeScript, Node.js/Express, and MongoDB that enables users to report, comment, and upvote human rights incidents.
 - Implemented RESTful API endpoints, user authentication, and form-based reporting with real-time database persistence.
 - Designed scalable data models for Users, Issues, Comments, and Interactions, ensuring strong type safety and reducing data inconsistencies by 30%.
- Wholesale Management System Database** | [Github Repo](#) | [Power BI](#) Nov 2023
- Designed a normalized MySQL schema and implemented stored procedures, triggers, and views to automate reporting workflows, reducing manual data processing by 50%.
 - Developed and integrated analytics dashboards in Power BI connected to live SQL queries, enabling managers to view key metrics in real time.
 - Optimized query execution plans, indexing strategies, and join operations to cut report generation time by 30%, significantly improving decision-making speed.

Technologies

Frontend: ReactJS, TypeScript, HTML5, CSS3, Bootstrap, ThreeJS
Backend: Node.js, ExpressJS, REST APIs
Databases: MongoDB, MySQL, PostgreSQL, MS SQL Server
Tools: GitHub, Postman, Vite, Power BI